Hardware Components

Core Processing Node

- **ESP32 Development Board:** The main processing unit handling analog-to-digital conversions and TLS 1.2 encryption.

Sensors & Display

- **DC Voltage Sensor:** Wired in parallel to measure the voltage potential of the load circuit.
- **ACS712 Current Sensor:** Wired in series to measure the physical current drawn.
- **16x2 LCD Display with I2C Backpack:** Provides local, real-time data viewing via the D21 and D22 pins.

The Isolated Load Circuit

- **DC Battery:** The physical power source driving the isolated high-voltage loop.
- **DC Fan:** The physical high-voltage load drawing power from the battery.

Power & Prototyping

- **DC005 Barrel Jack:** Provides the isolated 5V power and ground to the ESP32 and sensor logic side.
- **Standard Breadboard:** The physical mounting platform for your logic components.
- **Jumper Wires:** Connects the logic rails and bridges the high-voltage paths.

Infrastructure & Networking

- **POCO X7 Pro Smartphone:** Acts as your physical Middle Node, providing the isolated local area network hotspot.
- **Windows Laptop:** The physical Infrastructure Node hosting your Mosquitto broker and Möbius database.

Software Components

- **Custom C++ Firmware:** The application logic flashed to the ESP32. It handles sensor math, Wi-Fi connectivity, secure MQTT packet publishing, and oneM2M JSON payload structuring.
- **Möbius Resource Browser:** A Node.js web application (app.js) used as a graphical user interface to monitor and visualize the oneM2M resource tree in real-time.

- **OpenSSL Certificates:** The ca.crt, server.crt, and server.key files required to enable TLS 1.2 encryption on the Mosquitto broker.

Platforms and Services Used

Platforms & Servers

- **Möbius oneM2M Platform:** An enterprise-grade, open-source IoT server acting as the Common Services Entity (CSE). It manages access control, authenticates identity headers (SOrigin), and securely stores the sensor data in nested containers.
- **Eclipse Mosquitto:** An open-source message broker implementing the MQTT protocol. Configured as a dual-listener bridge to accept external encrypted MQTTS traffic (Port 8883) and route it to the internal Möbius database (Port 1883).
- **Blynk IoT Platform:** A cloud-based presentation layer platform used to build the mobile dashboard and display live energy metrics to the end user.

Development & Testing Tools

- **Arduino IDE:** The primary integrated development environment used to write, compile, and upload the C++ firmware to the ESP32 edge node.
- **Postman:** An API development tool used extensively to provision the oneM2M database structure (POST requests) and verify secure data retrieval (GET requests) using strict HTTP headers.
- **Wireshark:** A network protocol analyzer used during the testing phase to capture live network traffic and definitively prove the successful implementation of TLS 1.2 ciphertext encryption.
- **GitHub:** The version control service used to host the official Mar26_GrpName project repository, securely storing the final source code and server configuration files.